

Natural Language Processing: Question Answering

H02B1A-H02B1B



Miryam de Lhoneux

Slides adapted from Sien Moens

*Department of Computer Science
KU Leuven*

Outline

Introduction

Dimensions of QA

Pre-neural approaches to QA

Neural models and current benchmarks

Summary and outlook

Broader outlook

Practical info

What is Question Answering

Question answering

Question answering (QA) is the task of **answering questions** posed by humans in a natural language.

Question: What is the capital of Belgium?

Answer: Brussels

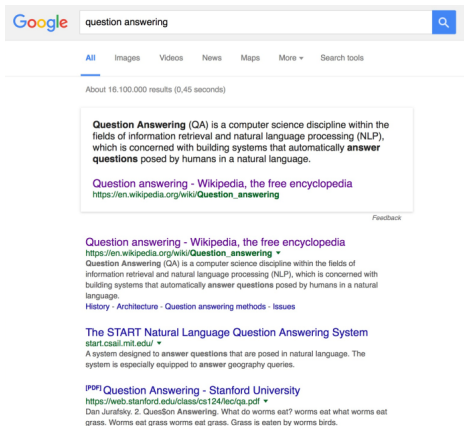
Formally

Answering questions is the problem of predicting an answer a given an image or other contextual source x and a question q :

$$\hat{a} = \arg \max_{a \in A} P(a|x, q)$$

Applications of QA

Classical search engines retrieve documents/web pages



Applications of QA

But users are lazy and want more automation:

- ✗ looking for the answer in the document/webpage
- ✓ answer pinpointed

Search and document retrieval

who developed the algorithm for the black hole image

Business Insider
Astronomers used AI to generate picture of black hole spotted in 2019
 A machine learning algorithm... researchers seems to be... clearer image of the M87 black hole... released...
 1 week ago

Sky & Telescope
The First Black Hole Image, Taken by the Event Horizon Telescope
 A new analysis of Event Horizon Telescope... the glowing gas encircling the black hole. AddThis Share...
 2 days ago

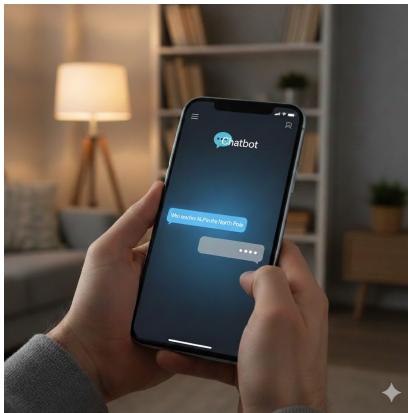
Scientific American
See the Sharp New Image of the Black Hole
 Scientific American is the essential guide to the most awe-inspiring advances in science and technology, explaining how they change our...
 1 week ago

Bouman joined Event Horizon Telescope project in 2013. He led the development of an algorithm for imaging black holes, known as Continuous High-resolution Image Reconstruction using Patch priors (CHIRP).

Wikipedia
https://en.wikipedia.org/wiki/Katie_Bouman
Katie Bouman - Wikipedia

Applications of QA

Dialogue systems: user input is typically questions



- QA is one of the earliest and core topics in natural language processing
- Plays a central role in many real-world applications such as search engines and personal assistants
- Is often used as test for language understanding

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Dimensions of QA

- Types of questions
- Types of information repositories
- Types of answers

Types of questions

1. Factual questions:

- *When did Mozart die?*
- Answer selected verbatim in text or generated

2. Questions that need **simple reasoning techniques**:

- *How did Socrates die?* → *die* has to be linked with *drinking poisoned wine*

3. Questions that need **answer fusion from different documents**:

- e.g., *In what countries did an earthquake occur last year?*

Types of questions

4. Interactive questions:

- Interaction of the user asking multiple questions
- Interaction of the system:

User What is the rotation time around the earth of a satellite?

System Which kind of satellite: GEO, MEO or LEO ?

5. Questions that need **analogical reasoning** e.g. *Is the US moving towards a recession?*:

No answer in data source. Need knowledge sources and temporal/spatial/evidential reasoning.

Types of repositories

- Database
- Knowledge base/knowledge graph
- Text archive
- Image repository
- Video archive
- Audio archive
- World Wide Web
- Combination of the above

Types of data in repositories

- **Structured**: stored objects and their attributes have well-defined semantics and relationships (e.g., relational database).
- **Unstructured**: information that is not organized in a pre-defined manner.

Closed or restricted domain question answering

Questioning a specific database.

Open domain question answering

Aims to automatically answer questions posed by humans on a large collection of unstructured documents.

Types of answers

- Structured values obtained from a database
- Phrasings, paragraphs or sentences that contain the answers
- Well-formed text answers
- Pictures, videos, ...
- Combination of the above

Types of answers for evaluation

What is considered a "correct" answer?

- A document, video containing the answer?
- A paragraph, sentence, image containing the answer?
- A word or phrase containing the answer?
- What if part of the answer is missing?

QA evaluation

- Accuracy: answer is correct or incorrect and mean over all questions
- WUPS score: like accuracy, but incorporates similarity between answer words using a threshold
- Precision: percentage of retrieved answers that are correct
- Recall: percentage of correct answers that have been retrieved
- Precision@1, Precision@5, ..., Recall@1, Recall@5, ...
- Mean reciprocal rank (MRR): based on the inverse of the rank at which the correct answer is found.

Mean Reciprocal Answer Rank (MRR)

- The top α answers for query k are checked and the score is discounted by the rank $r(k)$ at which it is found.
- $r(k) = 1$ if the first answer is relevant/correct, 2 if the second answer is relevant/correct,
- The scores across the n queries are then averaged:

$$MRR = \frac{1}{n} \left(\sum_{k=1}^n \frac{1}{r(k)} \right),$$

- If the answer is not found for a query, the score for that query k is 0.

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Early systems

- BASEBALL (Green et al., 1961): natural language interface to database with info on baseball games
Q: How many games did the Yankees play in July?
- LUNAR (Woods, 1973): natural language interface to database describing moon rock samples
Q: Give me all lunar samples with magnetite.
- SHRDLU (Winograd, 1972): interactive dialog interface to control robot interactions
Person: Pick up a big red block...

Limitations

- Operate on closed domain.
- Works only on well-controlled sublanguage.

1980s and 1990s: QA of knowledge bases built for restricted applications

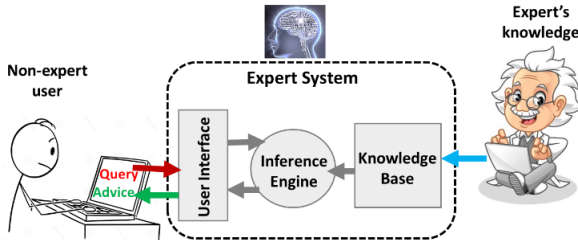
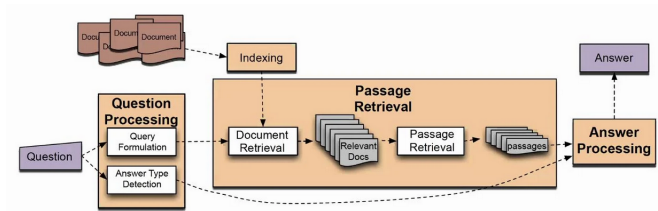


Figure from Katamoura and Aksoy (2022)

TREC

Yearly conference organised by NIST (national Institute of Standards and Technology, USA) with a QA track 1999-2007

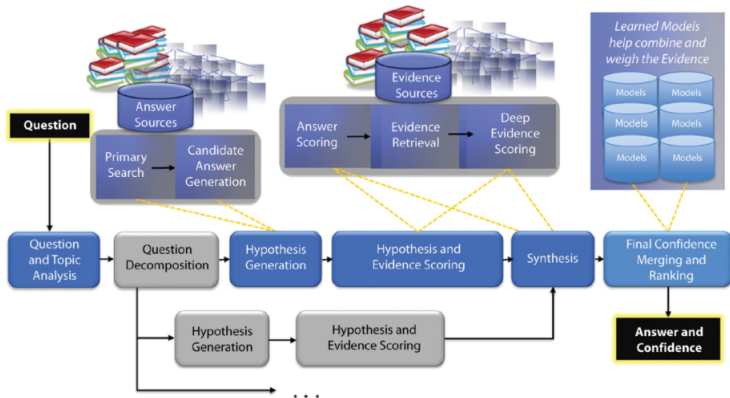


Typical system



IBM's Watson supercomputer beats humans in Jeopardy - 2011.

See also [link](#)



IBM Watson architecture (Ferrucci, 2012)

Query and document representations

- Bag-of-words representations
- Representations that incorporate a morpho-syntactic analysis
- Representations that incorporate a semantic analysis
- Representations that incorporate a translation to a knowledge representation

Bag-of-words representation

Vocabulary

{lawyer, car, blue, theft, family}

Doc1	a family buying a blue car	(0, 1, 1, 0, 1)
Doc2	a blue car theft defended by a lawyer	(1, 1, 1, 1, 0)
Doc3	a family asking a lawyer's advice	(1, 0, 0, 0, 1)
Doc4	a lawyer asking a family's advice	(1, 0, 0, 0, 1)

Query

a blue family car (0, 1, 1, 0, 1)

Bag-of-words representation

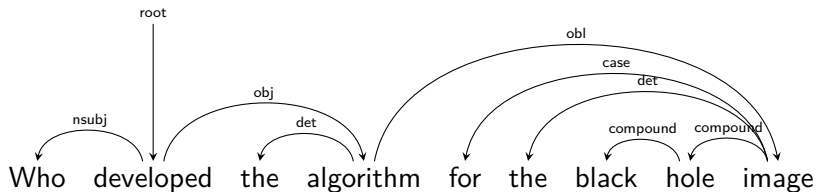
Vocabulary

{lawyer, car, blue, theft, family}

- Only consider words (possibly without stopwords and other non-content bearing words) without positional and grammatical information.
- Surprisingly effective if answer is redundant across the web

Representations that incorporate a morpho-syntactic analysis

- Lemmatisation (see lecture 7)
- Part-of-speech tagging (see lecture 6)
 - Resume|NOUN $\neq$ resume|VERB
- Syntactic analysis (see lecture 8)
 - Use of tree kernels or tree edit models for matching



Representations that incorporate a semantic analysis

Classification of the question or the expected answer type:

who/whom? PERSON

when? TIME/DATE

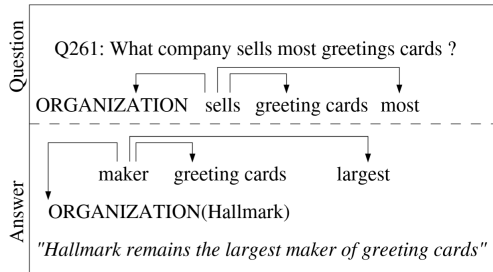
where/what place? LOCATION

what time (of day)? TIME

what/which (city - country) ? CITY - COUNTRY

Representations that incorporate a semantic analysis

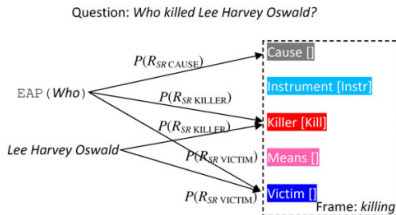
- NER (see lecture 6) on candidate answers
- Using WordNet (see lecture 9) to identify semantic relations (between *sell* and *maker*)



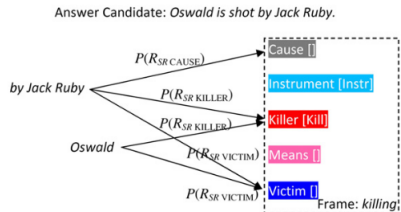
Matching question and answer aided by semantic analysis (Harabagiu et al., 2001)

Representations that incorporate a semantic analysis

- Semantic Role Labelling (see lecture 9) of question and candidate answers
- Imposing label matching



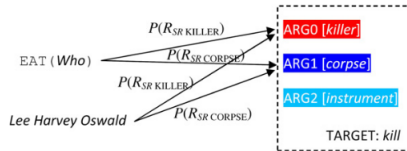
(a) Question Semantic Structure



(b) Answer Candidate Semantic Structure

FrameNet semantic role labelling representation (Kolomiyets and Moens, 2011)

Representations that incorporate a semantic analysis



PropBank semantic role labelling representation (Kolomiyets and Moens, 2011)

Representations that incorporate a semantic analysis

- If the semantic labels are accurate, good gain in performance.
- But this is not straightforward due to limited annotated data (see lecture 9).

Translation to a knowledge representation

- Translation of question to e.g., structured query language (SQL) allowing to interrogate relational database
- **Semantic parsing**: Translation of questions and context to logical forms, e.g.:

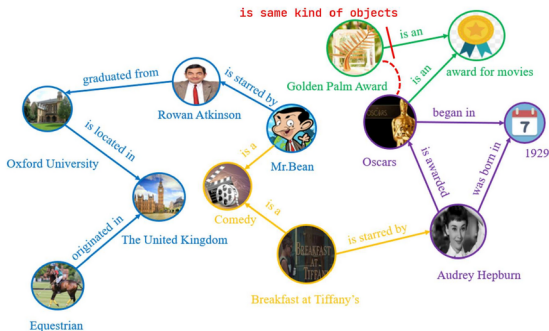
What is the capital of Texas?

$\lambda x. \text{city}(x) \wedge \text{capital}(x, \text{Texas})$

- This translation can be obtained by training a classifier on annotated pairs or converting a syntactic tree

Works well in closed domain settings.

QA over knowledge bases

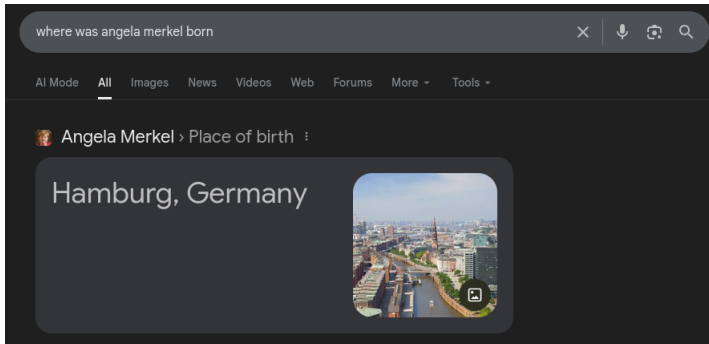


Example knowledge graph in Freebase (Bollacker et al., 2008). Figure from Jin and Yang (2023).

Question

From which university did Rowan Atkinson graduate?

QA over knowledge bases



Example question where wikidata (Vrandečić and Krötzsch, 2014) is used to find the answer.

QA over knowledge bases

Freebase: Structured ontology curated by domain experts
(discontinued, donated to Wikidata)

Wikidata: Community-driven open knowledge graph

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Neural representations

Neural models to represent questions and documents:

- Recurrent neural network such as LSTMs
- Recursive NNs based on dependency trees
- Attention-based models (transformers)

SQuAD

100k machine comprehension questions (Rajpurkar et al., 2016).

In meteorology, precipitation is any product of the condensation of atmospheric water vapor that falls under **gravity**. The main forms of precipitation include drizzle, rain, sleet, snow, **graupel** and hail... Precipitation forms as smaller droplets coalesce via collision with other rain drops or ice crystals **within a cloud**. Short, intense periods of rain in scattered locations are called "showers".

What causes precipitation to fall?

gravity

What is another main form of precipitation besides drizzle, rain, snow, sleet and hail?

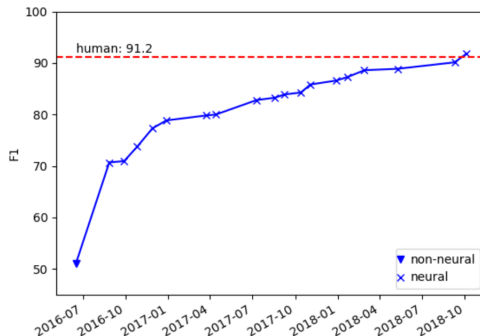
graupel

Where do water droplets collide with ice crystals to form precipitation?

within a cloud

SQuAD

100k machine comprehension questions (Rajpurkar et al., 2016).



SQuAD v2.0

SQuAD v1 + 50k unanswerable questions (Rajpurkar et al., 2018).

Article: Endangered Species Act

Paragraph: “... Other legislation followed, including the Migratory Bird Conservation Act of 1929, a 1937 treaty prohibiting the hunting of right and gray whales, and the Bald Eagle Protection Act of 1940. These later laws had a low cost to society—the species were relatively rare—and little opposition was raised.”

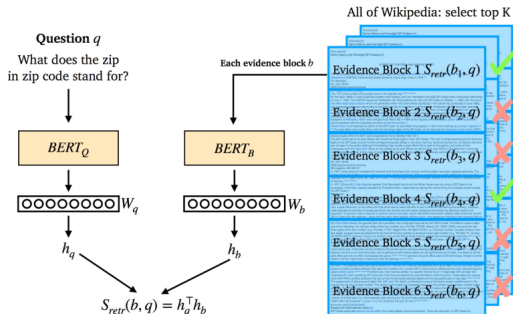
Question 1: “Which laws faced significant opposition?”

Plausible Answer: later laws

Question 2: “What was the name of the 1937 treaty?”

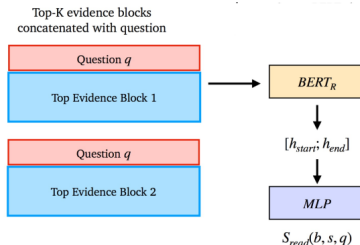
Plausible Answer: Bald Eagle Protection Act

Retrieving + reading comprehension



The retriever gives a score S_{retr} for a block b and a question q . The top- K candidate answer passages are retrieved. [Lee et al. \(2019\)](#).

Retrieving + reading comprehension



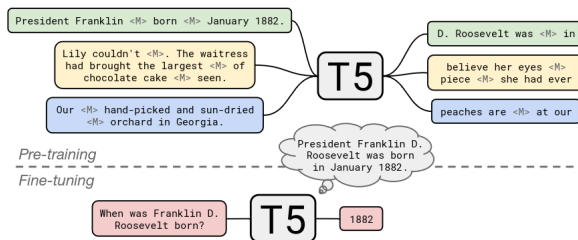
The reader gives a score S_{read} for a block b , an answer span s and a question q . Lee et al. (2019).

The model outputs the answer string a^* with the highest score:

$$a^* = \operatorname{argmax}_{b,s} S(b, s, q)$$

Retrieval-free QA

Can a pre-trained language model (e.g., trained on Wikipedia) act as a knowledge storage? **LM = KB**



T5 is pre-trained to fill in dropped-out spans of text (denoted by <M>) from documents in a large, unstructured text corpus. It is then fine-tuned to answer questions without inputting any additional information or context (Roberts et al., 2020).

Retrieval-free QA

Table 1: Scores achieved by fine-tuning T5 on the open-domain Natural Questions (NQ), WebQuestions (WQ), and TriviaQA (TQA) tasks.

	NQ	WQ	TQA	
			dev	test
Chen et al. (2017)	–	20.7	–	–
Lee et al. (2019)	33.3	36.4	47.1	–
Min et al. (2019a)	28.1	–	50.9	–
Min et al. (2019b)	31.8	31.6	55.4	–
Asai et al. (2019)	32.6	–	–	–
Ling et al. (2020)	–	–	35.7	–
Guu et al. (2020)	40.4	40.7	–	–
Férvy et al. (2020)	–	–	43.2	53.4
Karpukhin et al. (2020)	41.5	42.4	57.9	–
T5-Base	25.9	27.9	23.8	29.1
T5-Large	28.5	30.6	28.7	35.9
T5-3B	30.4	33.6	35.1	43.4
T5-11B	32.6	37.2	42.3	50.1
T5-11B + SSM	34.8	40.8	51.0	60.5
T5.1.1-Base	25.7	28.2	24.2	30.6
T5.1.1-Large	27.3	29.5	28.5	37.2
T5.1.1-XL	29.5	32.4	36.0	45.1
T5.1.1-XXL	32.8	35.6	42.9	52.5
T5.1.1-XXL + SSM	35.2	42.8	51.9	61.6

Comparison to baselines that use an external knowledge base (Roberts et al., 2020).

Retrieval-free QA

Setting	NaturalQS	WebQS	TriviaQA
RAG (Fine-tuned, Open-Domain) [LPP ⁺ 20]	44.5	45.5	68.0
T5-11B+SSM (Fine-tuned, Closed-Book) [RRS20]	36.6	44.7	60.5
T5-11B (Fine-tuned, Closed-Book)	34.5	37.4	50.1
GPT-3 Zero-Shot	14.6	14.4	64.3
GPT-3 One-Shot	23.0	25.3	68.0
GPT-3 Few-Shot	29.9	41.5	71.2

Fine-tuned T5 compared to not fine-tuning GPT-3 and RAG.

- GPT-3 is much larger than T5
- Retrieval-Augmented Generation (RAG) is the new wave of knowledge-based methods
 - Improves accuracy
 - Reduces hallucination
- More on this in lecture 12

Conversational QA

Jessica went to sit in her rocking chair. Today was her birthday and she was turning 80. Her granddaughter Annie was coming over in the afternoon and Jessica was very excited to see her. Her daughter Melanie and Melanie's husband Josh were coming as well. Jessica had ...

Q₁: Who had a birthday?

A₁: Jessica

R₁: Jessica went to sit in her rocking chair. Today was her birthday and she was turning 80.

Q₂: How old would she be?

A₂: 80

R₂: she was turning 80

Q₃: Did she plan to have any visitors?

A₃: Yes

R₃: Her granddaughter Annie was coming over

Q₄: How many?

A₄: Three

R₄: Her granddaughter Annie was coming over in the afternoon and Jessica was very excited to see her. Her daughter Melanie and Melanie's husband Josh were coming as well.

Q₅: Who? **Need dialogue history**

A₅: Annie, Melanie and Josh

R₅: Her granddaughter Annie was coming over in the afternoon and Jessica was very excited to see her. Her daughter Melanie and Melanie's husband Josh were coming as well.

Reddy et al. (2019)

Conversational QA

Jessica went to sit in her rocking chair. Today was her birthday and she was turning 80. Her granddaughter Annie was coming over in the afternoon and Jessica was very excited to see her. Her daughter Melanie and Melanie's husband Josh were coming as well. Jessica had ...

Q₁: Who had a birthday?

A₁: Jessica

R₁: Jessica went to sit in her rocking chair. Today was her birthday and she was turning 80.

Q₂: How old would she be?

A₂: 80

R₂: she was turning 80

Q₃: Did she plan to have any visitors?

A₃: Yes

R₃: Her granddaughter Annie was coming over

Q₄: How many? **No span-based answer**

A₄: Three

R₄: Her granddaughter Annie was coming over in the afternoon and Jessica was very excited to see her. Her daughter Melanie and Melanie's husband Josh were coming as well.

Q₅: Who?

A₅: Annie, Melanie and Josh

R₅: Her granddaughter Annie was coming over in the afternoon and Jessica was very excited to see her. Her daughter Melanie and Melanie's husband Josh were coming as well.

Reddy et al. (2019)

Conversational QA

Need for coreference resolution (see lecture 10).

Q₁: What are the candidates **running** for?

A₁: Governor

R₁: The Virginia governor's race

Q₂: **Where**?

A₂: Virginia

R₂: The Virginia governor's race

Q₃: Who is the democratic candidate?

A₃: **Terry McAuliffe**

R₃: Democrat Terry McAuliffe

Q₄: Who is **his** opponent?

A₄: **Ken Cuccinelli**

R₄: Republican Ken Cuccinelli

Q₅: What party does **he** belong to?

A₅: Republican

R₅: Republican Ken Cuccinelli

Q₆: Which of **them** is winning?

A₆: Terry McAuliffe

R₆: Democrat Terry McAuliffe, the longtime political fixer and moneyman, hasn't trailed in a poll since May

Reddy et al. (2019)

Conversational QA

CoQA

Need for

- Information collected through conversation (including coreference resolution)
- Pragmatic reasoning

Conversational QA

	Child.	Liter.	In-domain			Out-of-dom.		In-domain	Out-of-dom.	
			Mid-High.	News	Wiki.	Reddit	Science	Overall	Overall	Overall
Development data										
Seq2seq	30.6	26.7	28.3	26.3	26.1	N/A	N/A	27.5	N/A	27.5
PGNet	49.7	42.4	44.8	45.5	45.0	N/A	N/A	45.4	N/A	45.4
DrQA	52.4	52.6	51.4	56.8	60.3	N/A	N/A	54.7	N/A	54.7
Augmt. DrQA	67.0	63.2	63.9	69.8	72.0	N/A	N/A	67.2	N/A	67.2
DrQA+PGNet	64.5	62.0	63.8	68.0	72.6	N/A	N/A	66.2	N/A	66.2
Human	90.7	88.3	89.1	89.9	90.9	N/A	N/A	89.8	N/A	89.8
Test data										
Seq2seq	32.8	25.6	28.0	27.0	25.3	25.6	20.1	27.7	23.0	26.3
PGNet	49.0	43.3	47.5	47.5	45.1	38.6	38.1	46.4	38.3	44.1
DrQA	46.7	53.9	54.1	57.8	59.4	45.0	51.0	54.5	47.9	52.6
Augmt. DrQA	66.0	63.3	66.2	71.0	71.3	57.7	63.0	67.6	60.2	65.4
DrQA+PGNet	64.2	63.7	67.1	68.3	71.4	57.8	63.1	67.0	60.4	65.1
Human	90.2	88.4	89.8	88.6	89.9	86.7	88.1	89.4	87.4	88.8

Table 7: Models and human performance (F1 score) on the development and the test data.

Reddy et al. (2019)

Conversational QA

Leaderboard

Rank	Model	In-domain	Out-of-domain	Overall
	Human Performance <i>Stanford University</i> (Reddy & Chen et al. TACL '19)	89.4	87.4	88.8
1 Sep 05, 2019	RoBERTa + AT + KD (ensemble) <i>Zhuiyi Technology</i> https://arxiv.org/abs/1909.10772	91.4	89.2	90.7
1 Apr 22, 2020	TR-MT (ensemble) <i>WeChatAI</i>	91.5	88.8	90.7
2 Sep 05, 2019	RoBERTa + AT + KD (single model) <i>Zhuiyi Technology</i> https://arxiv.org/abs/1909.10772	90.9	89.2	90.4

Current state of the CoQA leaderboard.

Visual Question Answering (VQA)

Vehicles and Transportation



Q: What sort of vehicle uses this item?
A: firetruck

Brands, Companies and Products



Q: When was the soft drink company shown first created?
A: 1898

Objects, Material and Clothing



Q: What is the material used to make the vessels in this picture?
A: copper

Sports and Recreation



Q: What is the sports position of the man in the orange shirt?
A: goalie

Cooking and Food



Q: What is the name of the object used to eat this food?
A: chopsticks

Geography, History, Language and Culture



Q: What days might I most commonly go to this building?
A: Sunday

People and Everyday Life



Q: Is this photo from the 50's or the 90's?
A: 50's

Plants and Animals



Q: What phylum does this animal belong to?
A: chordate, chordata

Science and Technology



Q: How many chromosomes do these creatures have?
A: 23

Weather and Climate



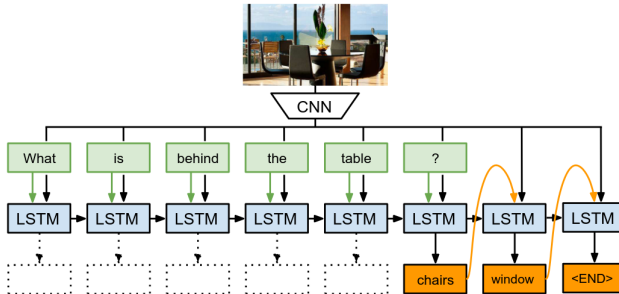
Q: What is the warmest outdoor temperature at which this kind of weather can happen?
A: 32 degrees

Examples from the open knowledge VQA (OKVQA) dataset (Marino et al., 2019)

VQA

VQA requires:

- Understanding of vision
- Understanding of language
- Commonsense knowledge

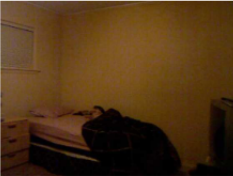
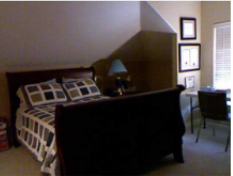


Example approach to a scene-based QA. [Malinowski et al. \(2015\)](#).

	Accu- racy	WUPS @0.9	WUPS @0.0
Average Consensus Metric			
Language only (ours)			
- multiple words	11.60	18.24	52.68
- single word	11.57	18.97	54.39
Neural-Image-QA (ours)			
- multiple words	11.31	18.62	53.21
- single word	13.51	21.36	58.03
Min Consensus Metric			
Language only (ours)			
- multiple words	22.14	29.43	66.88
- single word	22.56	30.93	69.82
Neural-Image-QA (ours)			
- multiple words	22.74	30.54	68.17
- single word	26.53	34.87	74.51

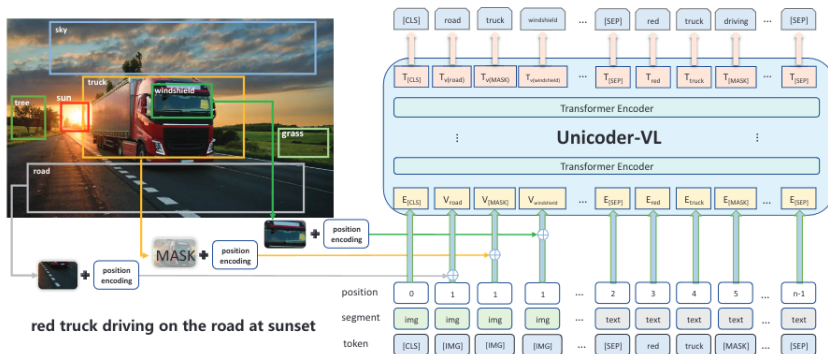
Table 5. Results on DAQUAR-Consensus, all classes, consensus in %.

Malinowski et al. (2015).

		
What is on the right side of the cabinet?	How many drawers are there?	What is the largest object?
<i>Neural-Image-QA:</i> bed	3	bed
<i>Language only:</i> bed	6	table

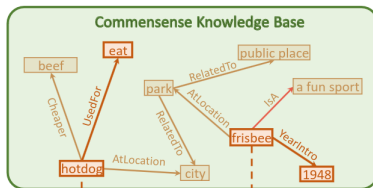
Examples of questions and answers. [Malinowski et al. \(2015\)](#).

State-of-the-art today: Vision-Language transformers (UNITER, VL-BERT, UNICODER-VL, etc.)



Unicoder-VL visualization (Li et al., 2019)

Knowledge-based VQA: requires common knowledge

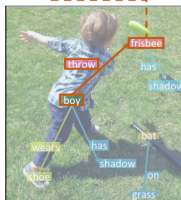


Q: What is the object that the girl holds used for?

A: Eat.

R: (girl, holds, hotdog)

(hotdog, UsedFor, eat)



Q: What is the relation between the boy and the object that first introduced in 1948 in the picture?

A: Throw.

R: (boy, throw, frisbee)

(frisbee, YearIntro, 1948)

KRVQR (Cao et al., 2020)

Outline

Introduction

Dimensions of QA

Pre-neural approaches to QA

Neural models and current benchmarks

Summary and outlook

Broader outlook

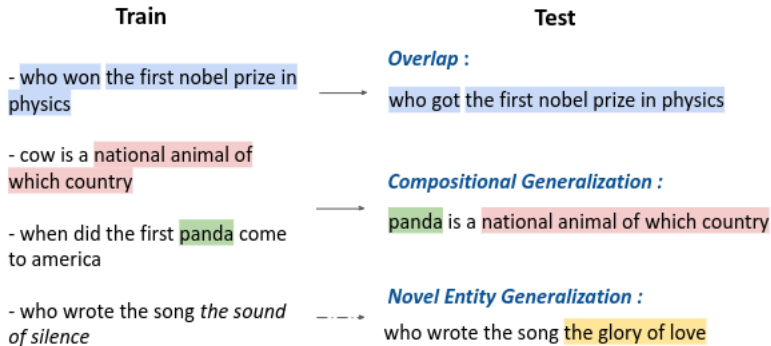
Practical info

QA evolution

- Rule-based to neural
- Complex pipelines to end-to-end to retrieval-free
- Closed-domain to open-domain
- Text-only to multimodal
- Large accuracy improvements over benchmarks, mainly thanks to transformers

Open challenges

- QA models memorize common question-answer relationships and still struggle with compositional generalization and new entities (Liu et al., 2022).
- A lot of research is driven by English
- A lot of non-English datasets are (machine-)translated from English
- There is a need for high quality multilingual datasets such as TyDi QA (Clark et al., 2020)
- These benchmarks tend to be more difficult!



(Liu et al., 2022)

Minimal Answer Task

Rank	Model	Participant	Affiliation	Attempt Date	F1	Precision	Recall
1	PrimeQA-Reader-with-Boolean	PrimeQA	PrimeQA team - IBM Research AI	11/1/2022	73.00	74.30	71.85
2	GAAMA-Syn-Bool-Single-Model	PrimeQA	PrimeQA team - IBM Research AI	9/7/2021	72.35	73.07	71.88
3	GAAMA-DM-Syn-ARES	PrimeQA	PrimeQA team - IBM Research AI	4/5/2021	68.06	74.38	62.92
4	PoolingFormer	PoolingFormer_Team	MSRA&Dynamics 365 AI	1/25/2021	67.65	73.48	63.27

TyDi QA leaderboard

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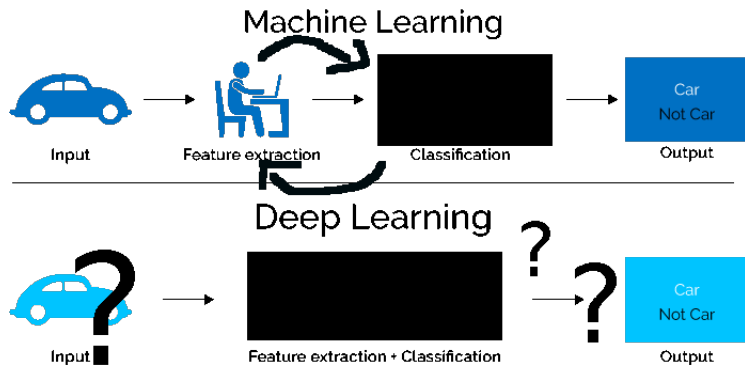
NLP evolution

- Rule-based (or symbolic) to probabilistic to neural
- Pipelines (morphology → syntax → semantics → discourse → downstream task) to end-to-end
- Are linguistic structures obsolete?

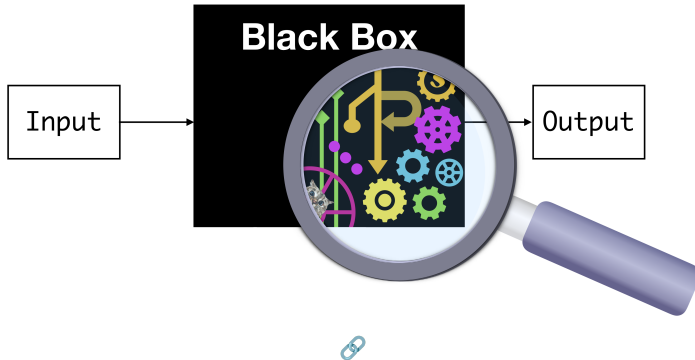
The state of machine learning



History of machine learning

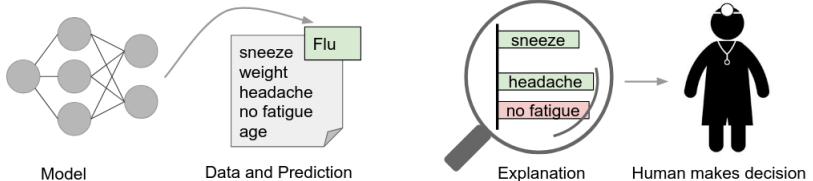


Interpretability



Interpretability: why?

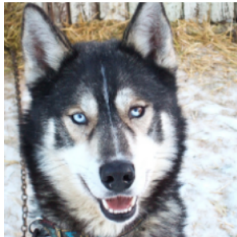
Make informed decisions



Ribeiro et al. (2016)

Interpretability: why?

- Get insights into errors
- Use the insights to improve the model



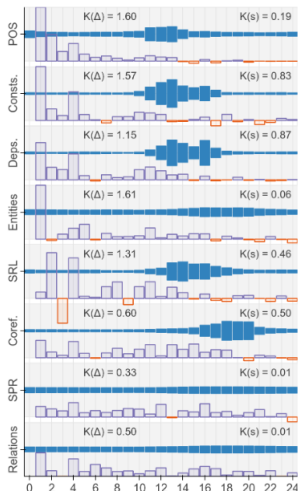
Husky classified as wolf



Explanation

Ribeiro et al. (2016)

Interpretability applied to NLP



BERT rediscovers the NLP pipeline (Tenney et al., 2019).

NLP evolution

- Ever larger number of parameters
- ⇒ Increasingly less interpretable models
- Linguistic structures are used to probe models
 - Provides insight in what they learn
 - Helps identify errors
- ⇒ Increasingly less ecologically sustainable models
- Linguistic structures may be part of more sustainable solutions (Neuro-Symbolic models)
 - No meaningful improvements for the majority of world languages
 - Linguistic structures may turn out to be useful for non high-resource languages

Outline

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Practical info

Rest of the semester

- Two more labs
- Next week (11/12): Q&A session
- One more invited lecture on 18/12 about conversational agents with GenAI by Thomas Winters
- Once the semester is over:
 - I will have limited capacity to answer emails
 - Do feel free to still use the course's forum

Q&A session of 11/12

What we will do

- We will go through the "mock" exam
 - Answer your questions about any lecture
 - ? Go through an exercise (again)
-
- Most useful with your input:
 - Attempt to answer the exam questions
 - Say what would you like to see (re-)explained?
 - See forums in the dedicated module on toledo
 - Your input so far:

Q&A session of 11/12

 Q&A session

- ☐  "Mock exam"
- ☐  Mock exam Q1
How would you answer the question?
- ☐  Mock exam Q2
How would you answer the question?
- ☐  Mock exam Q3
How would you answer the question?
- ☐  Mock exam Q4
How would you answer the question?
- ☐  Content questions
What are your content questions about any lecture of the course?
- ☐  Algorithms or exercises you found unclear
What are the algorithms or exercises you found unclear and you would like to be re-explained?

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